

ARDUINO CODE FOR RPMS-

```
#include <Wire.h>
```

```
#include <PulseOximeter.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
#define REPORTING_PERIOD_MS 1000
```

```
#define BUZZER_PIN 8
```

```
#define LM35_PIN A0
```

```
PulseOximeter pox;
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2); // Change to 0x3F if needed
```

```
uint32_t tsLastReport = 0;
```

```
void setup() {
```

```
    Serial.begin(115200);
```

```
    Wire.begin();
```

```
    pinMode(BUZZER_PIN, OUTPUT);
```

```
    digitalWrite(BUZZER_PIN, LOW);
```

```
    lcd.init();
```

```
    lcd.backlight();
```

```
    lcd.clear();
```

```
    lcd.setCursor(0,0);
```

```
    lcd.print("Initializing...");
```

```
    if (!pox.begin()) {
```

```
        lcd.clear();
```

```
        lcd.setCursor(0,0);
```

```
        lcd.print("MAX30100 ERROR");
```

```
while (1);  
}
```

```
pox.setIRLedCurrent(MAX30100_LED_CURR_7_6MA);
```

```
lcd.clear();  
lcd.print("System Ready");  
delay(1500);  
}
```

```
void loop() {  
  pox.update();  
  
  if (millis() - tsLastReport > REPORTING_PERIOD_MS) {
```

```
    float hr = pox.getHeartRate();  
    float spo2 = pox.getSpO2();
```

```
    int raw = analogRead(LM35_PIN);  
    float mv = raw * (5000.0 / 1023.0);  
    float tempC = mv / 10.0;
```

```
    Serial.print("HR: "); Serial.print(hr);  
    Serial.print(" SpO2: "); Serial.print(spo2);  
    Serial.print(" Temp: "); Serial.println(tempC);
```

```
    lcd.clear();  
    lcd.setCursor(0,0);  
    lcd.print("HR:");  
    if (hr > 0) lcd.print((int)hr);  
    else lcd.print("--");
```

```
lcd.print(" SpO2:");
```

```
if (spo2 > 0) lcd.print((int)spo2);
```

```
else lcd.print("--");
```

```
lcd.setCursor(0,1);
```

```
lcd.print("Temp:");
```

```
lcd.print(tempC,1);
```

```
lcd.print("C");
```

```
bool alarm = false;
```

```
if (hr > 120 || hr < 50) alarm = true;
```

```
if (spo2 < 92 && spo2 > 0) alarm = true;
```

```
if (tempC > 38.0) alarm = true;
```

```
if (alarm) digitalWrite(BUZZER_PIN, HIGH);
```

```
else digitalWrite(BUZZER_PIN, LOW);
```

```
tsLastReport = millis();
```

```
}
```

```
}
```